

Law of Large Numbers (LLN) & Central Limit Theorem (CLT)

1. Law of Large Numbers (LLN)

As the sample size increases, the sample average gets closer to the true population average (population mean).

2. Population and Samples

Population = entire dataset. Sample = subset of the population. Different samples have different means, but with larger sample sizes, these means get closer to the population mean.

3. Central Limit Theorem (CLT)

If we repeatedly draw sufficiently large random samples (usually $n \geq 30$), the distribution of sample means becomes approximately normal, even if the original population is not normally distributed.

4. Key Idea

For large samples: Sample Mean $\approx$ Population Mean and estimation error decreases.

5. LLN vs CLT

LLN tells us where the sample mean converges. CLT tells us the shape of the distribution of sample means.

6. Important Formulas

Population Mean = μ

Sample Mean = $\bar{x}$

Standard Error = $\sigma/\sqrt{n}$

7. Why CLT Matters

Used to build confidence intervals, perform hypothesis testing, estimate population parameters, and supports many ML/statistical methods.

8. Interview Tips

- Larger sample size reduces sampling error.
- CLT is about sample means, not raw data.
- Standard Deviation measures spread of data; Standard Error measures spread of sample means.
- Bigger samples improve accuracy but do not remove bias.
- Random and independent sampling is assumed.

9. Common Interview Questions

Q: Difference between LLN and CLT?

A: LLN explains convergence of the sample mean to the population mean; CLT explains why sample means follow a normal distribution.

Q: Why is $n \approx 30$ commonly used?

A: It is a practical rule of thumb where the normal approximation is often good enough.